

Smart Private Parking Platform

QR Check-in/Check-out + Hourly Billing + Razorpay Payment Integration

1. QR-Based Check-in & Check-out

Each confirmed parking booking generates a unique booking ID and QR code. The QR code is used to authenticate the user's booking at the parking location.

Check-in: User scans the QR code → backend validates the booking → server records the check-in timestamp.

Check-out: User scans the QR code again → backend records the check-out timestamp → duration and any extra charge are calculated.

Important: The QR code does not itself run the timer. The backend stores the server-side check-in/check-out timestamps.

2. Hourly Booking & Billing Model

Scenario	System Behaviour
User books 3 hours	User pays the booking amount upfront.
User leaves early	Parking becomes available after checkout. For the MVP, no additional charge is required.
User stays beyond booked time	System calculates extra time and charges the applicable hourly rate.
Example	■20/hour × 3 hours = ■60 booked. Staying 1 extra hour → ■20 additional charge.

Recommended MVP: Use fixed upfront booking for the selected duration. Later, partial refunds for early checkout can be added if required.

3. Booking Data Stored by the Backend

A booking record should contain: **booking_id**, **user_id**, **parking_id**, **scheduled_start**, **scheduled_end**, **check_in_time**, **check_out_time**, **amount_paid**, **extra_amount**, **payment_status**, and **booking_status**.

4. Razorpay Payment Gateway

Razorpay can be integrated into the platform to collect online parking payments. The normal flow is:

User selects parking & duration → Backend creates booking/order → Razorpay Checkout → User completes payment → Backend verifies payment → Booking becomes CONFIRMED → QR code is generated.

Cost model: Standard payment gateway usage generally has no setup or annual maintenance fee, but successful real transactions are subject to Razorpay's applicable transaction charges. For development, Razorpay Test Mode can be used without real-money transactions.

5. Complete End-to-End Flow

1	Select parking location
2	Select date/time or duration
3	Create booking
4	Pay through Razorpay
5	Verify payment on backend

6	Generate booking QR
7	Scan QR → Check-in timestamp recorded
8	Scan QR → Check-out timestamp recorded
9	Calculate extra time/charge if applicable
10	Complete booking and release parking slot

6. Suggested MVP Technology

Component	Suggested Technology
Frontend	React / Next.js
Maps	Mapbox or Google Maps
Backend	Node.js / FastAPI
Database	PostgreSQL
Payment	Razorpay
Access / Usage	QR Code + server-side timestamps

Note: For production deployment, payment verification, refunds, marketplace/owner payouts, fraud prevention, privacy, and applicable legal/KYC requirements should be implemented according to the chosen payment provider and business model.